

1 Acquisition

views per plant
ordering (video or
discrete views)
pose, depth,
baseline, sides

2 Observations

detections or
masks in each
image; one
fruit may yield
several boxes

3 Association

decide which
observations belong
to the same fruit
(mechanisms
M0–M5 below)

4 Attributes

class label, size,
mass assigned to
the unique
instance, not to
each box

5 Inventory

one record per
physical fruit,
aggregated by
class, tree, row,
or block

M0 No association

Assumes:

each fruit is seen
once, or duplicates
are ignored

M1 Statistical correction

Assumes:

duplication is stable
enough to calibrate
against hand counts

M2 Appearance matching

Assumes:

fruit look different
enough to be matched
by appearance

M3 Temporal tracking

Assumes:

consecutive frames
overlap; motion is
smooth and bounded

M4 Geometric or 3D association

Assumes:

pose, scale, and
depth or reconstruction
are reliable

M5 Learned association

Assumes:

linked identities
exist to train and
test the matcher